A bio- and chemoinformatic approach for anthelmintic discovery in finfish aquaculture

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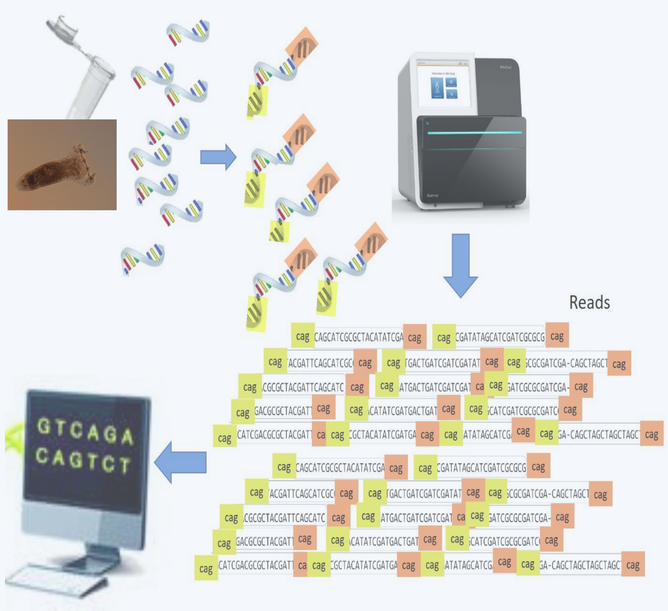
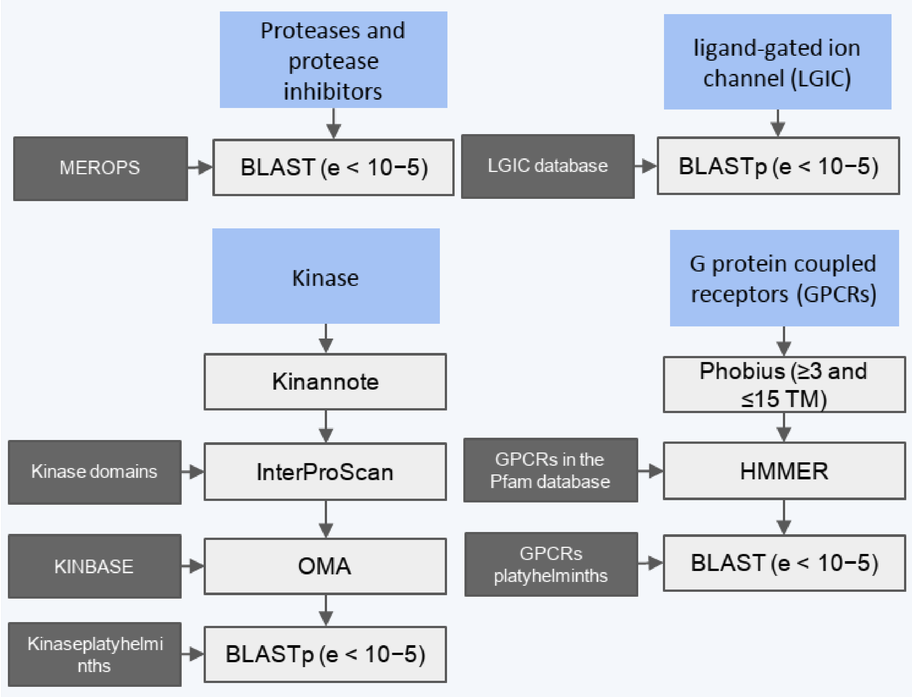
Introduction

The production of fish in aquaculture can be affected by monogeneans, which are parasitic flatworms (Ogawa, 2015). The drugs available to control these parasites are few and not always effective. Currently, computational studies of omic information have helped to propose certain groups of proteins as possible pharmacological targets (Rana, y Misra-Bhattacharya, 2013). This work presents a methodological framework based on the monogenean *Rhabdosynochus* sp. in order to identify drug targets and to identify drugs with affinity to the identified proteins. Iinitially proposed to generate information transcriptomics of the monogenean parasite *Rhabdosynochus* sp. that infects snook (*C. viridis)*. Subsequently, the genes that can be used as drug targets will be identified *in silico*. These genes will be used to identify FDA-approved drugs that can act as anthelmintics, using computational drug repositioning analysis. Finally, we will evaluate the anthelmintic activity with an in vitro phenotypic screening.

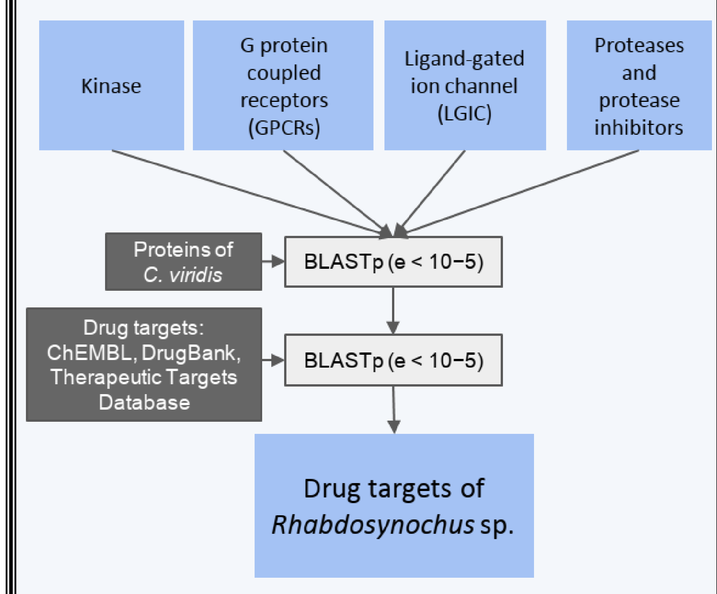
1. Transcriptome sequencing



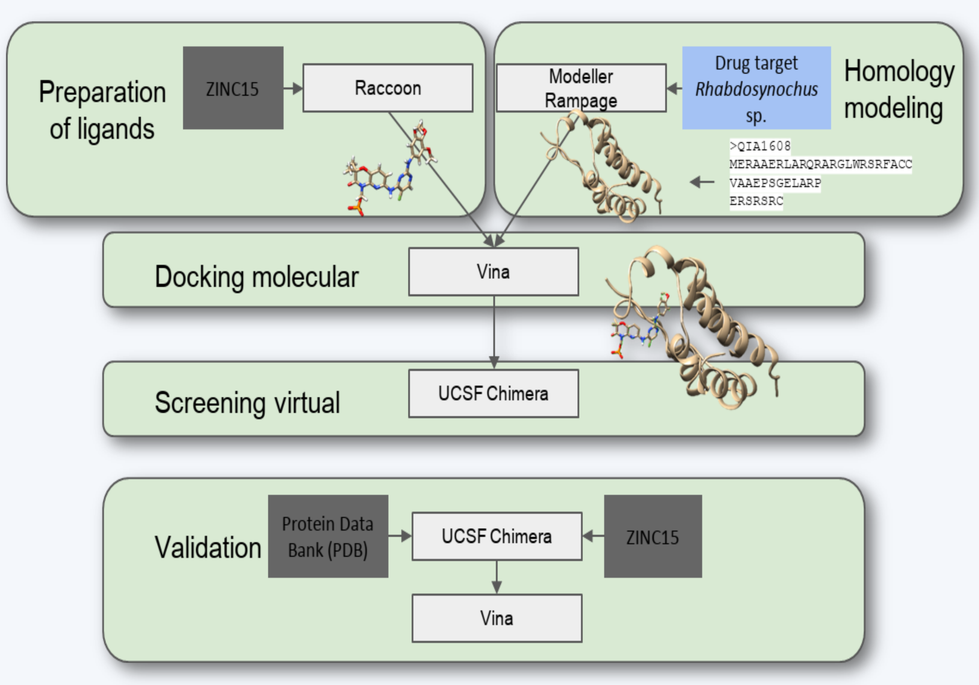
2. Identification of essential proteins



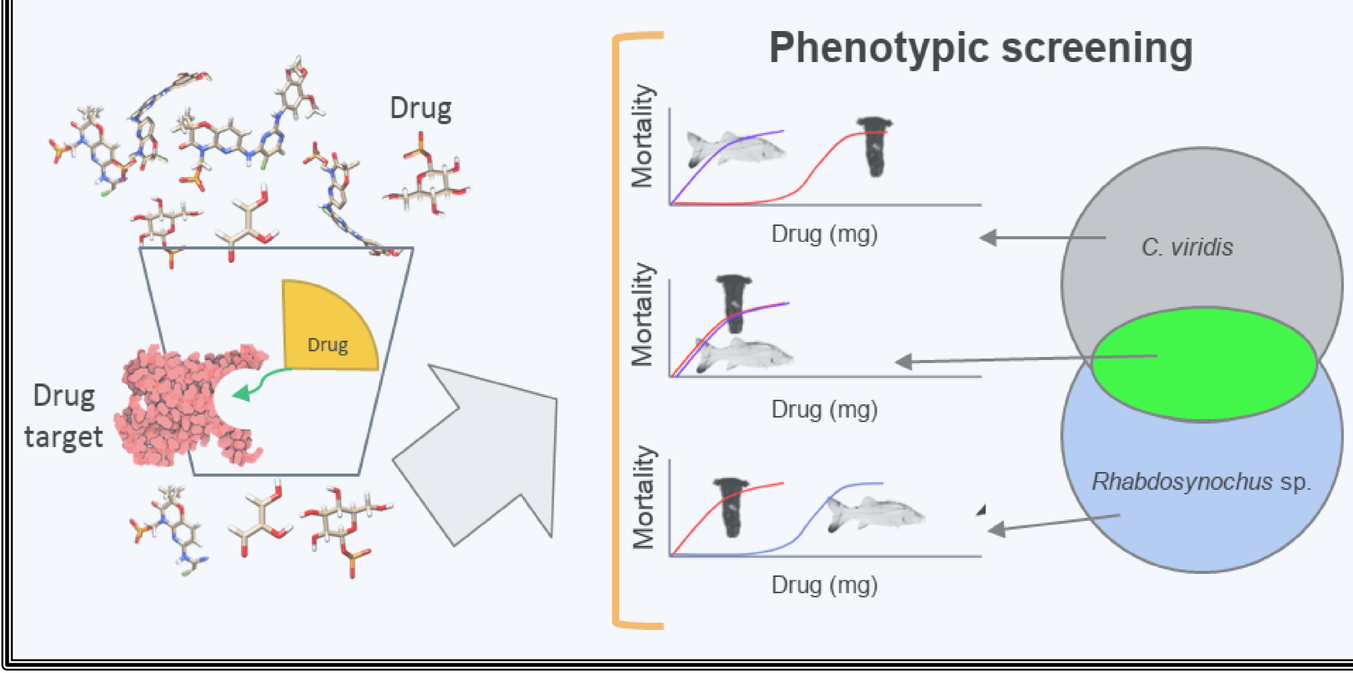
3. Identification of drug targets



4. Drug identification



5. Evaluation of anthelmintic activity



Conclusion

It is expected to be able to identify at least one drug target and reposition at least three drugs for each target drug and can evaluate the anthelmintic activity in in vitro and in vivo tests.

References

Ogawa K. 2015. Diseases of cultured marine fishes caused by Platyhelminthes (Monogenea, Digenea, Cestoda). Parasitology, 142(1):178-195.

Rana, A. K., & Misra-Bhattacharya, S. 2013. Current drug targets for helminthic diseases. Parasitology research, 112(5), 1819-1831.